Dr. Stavros Eleftherakis

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CV Summary: Postdoctoral Researcher at NYU Abu Dhabi specializing in wireless security, positioning systems, and AI-driven network analysis. PhD in Telematics Engineering with expertise spanning WiFi and 5G positioning, privacy in cellular systems, and secure wireless infrastructure. First author of publications in leading venues including ACM MobiCom and IEEE TMC, with research translated into an Ericsson industrial patent. Experienced in academia-industry collaboration across Europe and the Middle East.
Research Interests: Secure Positioning Systems, Wireless Security, 5G/6G and Non-Terrestrial Networks, Privacy in Cellular Systems, Applied Machine Learning for Wireless Networks

Education

09/2012-03/2017: University of Crete, Dep. of Mathematics and Applied Mathematics, Heraklion, Greece
- Diploma of Mathematics

09/2017-03/2020: University of Crete, Dep. of Mathematics and Applied Mathematics, Heraklion, Greece
- MSc Degree in Applied and Computational Mathematics (Supervisor: Dr. Georgios Kosioris)
- Master Thesis: Mathematical Techniques on Machine Learning

09/2020-09/2021: Universidad Carlos III de Madrid, Dep. of Telematics Engineering, Madrid, Spain
- MSc Degree in Telematics Engineering (Supervisor: Dr. Domenico Giustiniano)
- Master Thesis: Single Access Point Localization System

04/2022-09/2025: Universidad Carlos III de Madrid, Dep. of Telematics Engineering, Madrid, Spain
- PhD in Telematics Engineering (Supervisor: Dr. Domenico Giustiniano)
- PhD Thesis: Positioning Systems and the Challenges of Security & Privacy for Real-World Wireless Network Deployments

Professional Experience

07/2020-12/2025: IMDEA Networks Institute, Madrid, Spain
- Research Assistant in Pervasive Wireless Systems Group (Supervisor: Dr. Domenico Giustiniano)
- Description: Research on WiFi Positioning (IEEE TMC) & 5G Positioning (IEEE MAIN)

09/2023-03/2024: Telefonica Research, Barcelona, Spain
- PhD Intern (Supervisor: Dr. Nicolas Kourtellis)
- Description: Conducted research on privacy vulnerabilities in 5G Standalone networks, leading to first-author publications at ACM MobiCom, and ACM WiSec

06/2024-11/2024: Ericsson Research, Linköping, Sweden
- PhD Intern (Manager: Dr. Nicklas Johansson, Supervisor: Dr. Yuxin Zhao)
- Description: Co-invented Ericsson patent on integrity monitoring for 5G Timing Advance (TA)

01/2026 (current): New York University Abu Dhabi, Division of Science, Abu Dhabi, UAE
- Postdoctoral Associate in Cyber Security and Privacy Lab (Supervisor: Dr. Christina Pöpper)
- Description: Research on secure positioning, spoofing detection and wireless security including NTN

Publications

- **S. Eleftherakis**, G. Santaromita, M. Rea, T. Otim, D. Giustiniano, Contact Tracing Through Multipath Profile Similarity, ACM CoNEXT Workshop, Rome, Italy, 2022
- **S. Eleftherakis**, G. Santaromita, M. Rea, J. Costa-Perez, D. Giustiniano, SPRING+: Smartphone Positioning from a Single WiFi Access Point, **IEEE Transactions on Mobile Computing**, 2024
- **S. Eleftherakis**, T. Otim, G. Santaromita, A. Diaz-Zayas, D. Giustiniano, N. Kourtellis, Demystifying Privacy in 5G Stand Alone Networks, **ACM MobiCom**, Washington D.C., DC, USA, 2024
- **S. Eleftherakis**, D. Giustiniano, N. Kourtellis, SoK: Evaluating 5G-Advanced Protocols Against Legacy and Emerging Privacy and Security Attacks, **ACM WiSec**, Arlington, Virginia, USA, 2025
- **S. Eleftherakis**, Y. Zhao, X. Jiang, G. Lindmark, F. Gunnarsson, D. Giustiniano, From 5G Single-Cell Signals to Pervasive Smartphone Positioning: A Real-World Study, IEEE MAIN, Palermo, Italy, 2026

Patents

- **S. Eleftherakis**, Y. Zhao, X. Jiang, G. Lindmark, F. Gunnarsson, D. Giustiniano, T. Johansson, A Method for Timing Advance Integrity Monitoring, P112699GR01, **granted** December 2025 (Ericsson).

Programming Skills

- **Programming:** Python, MATLAB
- **Libraries/Frameworks:** TensorFlow, Keras, scikit-learn, NumPy, SciPy, Pandas, Matplotlib

Languages

- Greek: Native Language
- English: Fluent (C2, Michigan Proficiency)
- Spanish: Basic (A2)

Teaching Experience

- Teaching Assistant (2013–2020), University of Crete
- Courses: Differential Equations, Calculus I, Probability Theory, Applied Statistics, Analytical Geometry

Honours and Awards

- Manassaki Scholarship (University of Crete, 2019) for his high performance as a graduate student
- Travel Grant for ACM CoNEXT Conference, 2022
- Travel Grant for ACM MobiCom Conference, 2024

Invited Talks

- "Single Access Point Localization System", IMDEA Networks Weekly Seminar, Madrid, Spain, December 2021
- "Demystifying Privacy and Security in 5G Networks", Telefónica Research Weekly Seminar, Barcelona, Spain, June 2024 (online)
- "Cybersecurity and Citizen Science: Building Digital Resilience Together", TALOS Lab, Research Center for the Humanities, the Social and Education Sciences, University of Crete, Rethymnon, Greece, May 2026
- "What Can You Do with a Mathematics Degree? My Journey into Positioning Systems and Cybersecurity", Department of Mathematics & Applied Mathematics, University of Crete, Heraklion, Greece, May 2026
- "Measuring Real-World 5G Networks: From Single-Cell Positioning to Security and Privacy", MENA Cybersecurity Seminar Series, New York University Abu Dhabi, Abu Dhabi, UAE, June 2026

Academic Service

- **Technical Program Committee (TPC)**
  - Network and Distributed System Security (NDSS) Symposium 2027
  - 1st Workshop on Positioning Technologies for 5G & 6G (POSIT-6G), collocated with VTC2024-Fall
  - 16th ACM Wireless of the Students, by the Students, and for the Students Workshop, collocated with ACM MobiCom 2025

- **External Reviewer:**
  - Conferences: IEEE INFOCOM 2025, WCNC 2025
  - Journals: IEEE Transactions on Mobile Computing (Certified Reviewer, 2025), IEEE Transactions on Network Science and Engineering, Elsevier Computer Communications